1 Technical proofs

1.1 Sketch of Proof of Lemma 1

Let $z(t) := [z_1(t); \cdots ; z_M(t)] \in \mathbb{R}^{Mp}$, and with slight abuse of notation, let us define

$$\mathsf{Q}(z(t)) := [\mathsf{Q}(z_1(t); \cdots ; \mathsf{Q}(z_M(t))] \in \mathbb{Z}^{Mp}.$$

Then, (8) can be rewritten in vector form, as

$$z(t+1) = (W \otimes I_p)z(t) + ((W - I_M) \otimes I_p)(\mathsf{L}_z \cdot \mathsf{Q}(z(t)) - z(t)), \tag{1}$$

where each entry of the matrix $W = [W_{ij}] \in \mathbb{R}^{M \times M}$ is given by

$$W_{ij} := \begin{cases} w_{ij}, & j \in \mathcal{N}_i \\ 1 - \sum_{k \in \mathcal{N}_i} w_{ik}, & j = i \\ 0, & \text{otherwise} \end{cases}$$

for all $i \in \mathcal{V}$. It can be easily verified that (4) can also be rewritten as $z(t+1) = (W \otimes I_p)z(t)$. Thus, considering (1) in the supplementary material as a perturbed version of (4) with the perturbation term $((W - I_M) \otimes I_p)(\mathsf{L}_z \cdot \mathsf{Q}(z(t)) - z(t))$, the proof narrows down to analyzing the effect of the perturbation term on the convergence.

Since $\|\mathsf{L}_z \cdot \mathsf{Q}(z(t)) - z(t)\| \leq \mathsf{L}_z/2$ for all $t \geq 0$, the effect of the perturbation term can be made arbitrarily small by decreasing L_z. Also, the requirement for $C \geq 2$ ensures that the diagonal entries of W are dominant, i.e., $w_{ii} > 1/2$ and $\sum_{j \in \mathcal{N}_i} w_{ij} \leq 1/2$ for all $i \in \mathcal{V}$. This implies that increasing C decreases the induced norm of $W - I_M$, which further reduces the effect of the perturbation term. We refer the interested reader to (El Chamie, Liu, and Başar (2016)) for more details.

1.2 Proof of Lemma 2

It directly follows from (15c) that

$$\begin{aligned} \phi_{ii}(t) + \sum_{j \in \mathcal{N}_i} \phi_{ij}(t) \bmod q &= \sum_{j \in \mathcal{N}_i^+} \sum_{l \in \mathcal{N}_i^+ \cap \mathcal{N}_j^+} s^i_{l \to j}(t) \bmod q \\ &= \sum_{j \in \mathcal{N}_i^+} \sum_{l \in \mathcal{N}_i^+ \cap \mathcal{N}_j^+} s^i_{j \to l}(t) \bmod q, \end{aligned}$$

where the second equality holds because $j \in \mathcal{V}$ and $l \in \mathcal{V}$ satisfy $j \in \mathcal{N}_i^+$ and $l \in \mathcal{N}_i^+ \cap \mathcal{N}_j^+$ if and only if $l \in \mathcal{N}_i^+$ and $j \in \mathcal{N}_i^+ \cap \mathcal{N}_l^+$. Since

$$\sum_{l \in \mathcal{N}_i^+ \cap \mathcal{N}_l^+} s^i_{j \to l}(t) \bmod q = 0, \quad \forall j \in \mathcal{N}_i^+,$$

due to (15a) and (15b), this concludes the proof.

1.3 Proof of Theorem 1

We focus on the case of $T = 1$ because Protocol 1 with arbitrary $T \in \mathbb{N}$ can be realized as a sequential composition of Protocol 1 with $T = 1$, and such composition preserves security (Canetti (2000)), accordingly we have $\mathsf{y}_i = z_i(1)$.

The message received by agent i during the execution of the protocol is given by

$$\mathsf{msg}_i = (\{s^i_{j\to i}(0)\}_{j\in\mathcal{N}_i^+},\ \{\zeta_{ij}(0)\}_{j\in\mathcal{N}_i},\ \{s^j_{l\to i}\}_{j\in\mathcal{N}_i, l\in\mathcal{N}_i^+\cap\mathcal{N}_j^+}).$$

The first two components of msg_i correspond to the shares and masked message that agent i receives as an aggregator, while the last component represents the shares it receives when its neighbor is the aggregator.

Each share and masked message composing msg_i appears uniformly random in $\mathbb{Z}_q^p$, due to the randomness of the secret sharing algorithm and Assumption 1. However, it is important to note that they should satisfy the equalities given by (12) and (15c). In this regard, we construct a simulator Sim_i that outputs

$$\mathsf{Sim}_i(\mathsf{x}_i, \mathsf{y}_i) = (\mathsf{x}_i, (\{\hat{s}^i_{j\to i}(0)\}_{j\in\mathcal{N}_i^+}, \{\hat{\zeta}_{ij}(0)\}_{j\in\mathcal{N}_i}, \{\hat{s}^j_{l\to i}\}_{j\in\mathcal{N}_i, l\in\mathcal{N}_i^+\cap\mathcal{N}_j^+})),$$

where each $\hat{s}^i_{j\to i}(0)$, $\hat{\zeta}_{ij}(0)$, and $\hat{s}^j_{l\to i}$ appear uniformly random in $\mathbb{Z}_q^p$, and satisfy

$$\begin{aligned}\mathsf{y}_i &= z_i^{\mathsf{ini}} + \mathsf{L}_w\mathsf{L}_z\left(\hat{\phi}_{ii}(0) + \sum_{j\in\mathcal{N}_i}\hat{\zeta}_{ij}(0) - \bar{w}_{ij}\mathsf{Q}(z_i^{\mathsf{ini}}) \mod q\right)\\ \hat{\phi}_{ii}(0) &= \sum_{l\in\mathcal{N}_i^+}\hat{s}^i_{l\to i}(0) \mod q,\end{aligned} \tag{2}$$

where L_w, L_z, and q are publicly known. The simulator emulates as follows:

1. Sample $\hat{\zeta}_{ij}(0) \in \mathbb{Z}_q^p$ uniformly at random from $\mathbb{Z}_q^p$ for all $j \in \mathcal{N}_i$.
2. Sample $\hat{s}^j_{l\to i}(0) \in \mathbb{Z}_q^p$ uniformly at random from $\mathbb{Z}_q^p$ for all $j \in \mathcal{N}_i$ and $l \in \mathcal{N}_i^+ \cap \mathcal{N}_j^+$.
3. Using $(\mathsf{x}_i, \mathsf{y}_i)$, compute $\hat{\phi}_{ii}(0) \in \mathbb{Z}_q^p$ that satisfies the first equality of (2) in the supplementary material. Such $\hat{\phi}_{ii}(0)$ exists given that (14) holds.
4. Generate $(\hat{s}^i_{j\to i}(0))_{j\in\mathcal{N}_i^+} \leftarrow \mathsf{Share}(\hat{\phi}_{ii}(0), |\mathcal{N}_i^+|)$.

By construction, it is clear that $(\mathsf{x}_i, \mathsf{msg}_i) \equiv \mathsf{Sim}_i(\mathsf{x}_i, \mathsf{y}_i)$, and this concludes the proof.

1.4 Proof of Theorem 2

By Proposition 1, the output $z_i(T)$ of Protocol 2 with (18) satisfies

$$\limsup_{T\to\infty}\left\|\begin{bmatrix}z_{i,1}(T)\\ z_{i,2}(T)\end{bmatrix} - \begin{bmatrix}\sum_{i=1}^M V_i^{-1}(x)\cdot\hat{f}_i(x)\\ \sum_{i=1}^M V_i^{-1}(x)\end{bmatrix}\right\| \le \alpha(\mathsf{L}_z), \tag{3}$$

for all $i \in \mathcal{V}$. Hence, there exists $T_0 \ge 0$ such that

$$z_{i,2}(T) \ge \xi - \alpha(\mathsf{L}_z) > 0$$

for all $T \geq T_0$. For such T, it holds that

$$\left\| V_i^{(T)}(x) - V(x) \right\| = \left\| \frac{1}{z_{i,2}(T)} - \frac{1}{\xi} \right\| = \left\| \frac{\xi - z_{i,2}(T)}{z_{i,2}(T) \cdot \xi} \right\| \leq \frac{\alpha(\mathsf{L}_z)}{(\xi - \alpha(\mathsf{L}_z))\xi} =: \gamma(\mathsf{L}_z),$$

where γ is a class $\mathcal{K}$ function that satisfies (21), under the condition that $\xi > \alpha(\mathsf{L}_z)$.

To derive β, observe that there exists $T_1 \geq 0$ such that

$$\|z_{i,1}(T)\| \leq \left\| \sum_{i=1}^{M} V_i^{-1}(x) \cdot \hat{f}_i(x) \right\| + \alpha(\mathsf{L}_z)$$

for all $T \geq T_1$ by (3) in the supplementary material. Then, for such T, it follows that

$$\begin{aligned}
\left\| \hat{f}_i^{(T)}(x) - \hat{f}(x) \right\| &= \Bigg\| \left(V_i^{(T)}(x) - V(x) \right) z_{i,1}(T) \\
&\quad + V(x) \left(z_{i,1}(T) - \sum_{i=1}^{M} V_i^{-1}(x) \cdot \hat{f}_i(x) \right) \Bigg\| \\
&\leq \gamma(\mathsf{L}_z) \cdot \left(\left\| \sum_{i=1}^{M} V_i^{-1}(x) \cdot \hat{f}_i(x) \right\| + \alpha(\mathsf{L}_z) \right) + \frac{\alpha(\mathsf{L}_z)}{\xi} \\
&=: \beta(\mathsf{L}_z),
\end{aligned}$$

where we used the triangle inequality and (21). The function β is clearly of class $\mathcal{K}$, and this concludes the proof.

2 Additional Plots for Section 5.1

Let us define

$$\widehat{\mathsf{LMM}}(t) := \sum_{i=1}^{M} \log p(\mathcal{D}_i \mid \Theta_i(t)),$$

which approximates the the right-hand-side of (23) as the sum of the local log marginal likelihoods computed with the estimates $\Theta_i(t)$. It can be seen that $\widehat{\mathsf{LMM}}(t)$ converges to (23) as $\Theta_i(t)$ reach a consensus.

The two figures below illustrate the evolution of $\widehat{\mathsf{LMM}}(t)$ during the hyperparameter optimization procedure performed in Section 5.1, for both the SARCOS and Diabetes dataset. The results indicate that $\widehat{\mathsf{LMM}}(t)$ tends to monotonically increase over iterations, which is consistent with the objective of hyperparameter optimization. Optimization performance can be enhanced through careful tuning of the initial hyperparameter estimates and learning rate.

References

Canetti, R. 2000. Security and composition of multiparty cryptographic protocols. *J. Cryptol.*, 13: 143–202.

El Chamie, M.; Liu, J.; and Başar, T. 2016. Design and analysis of distributed averaging with quantized communication. *IEEE Trans. Autom. Control*, 61(12): 3870–3884.

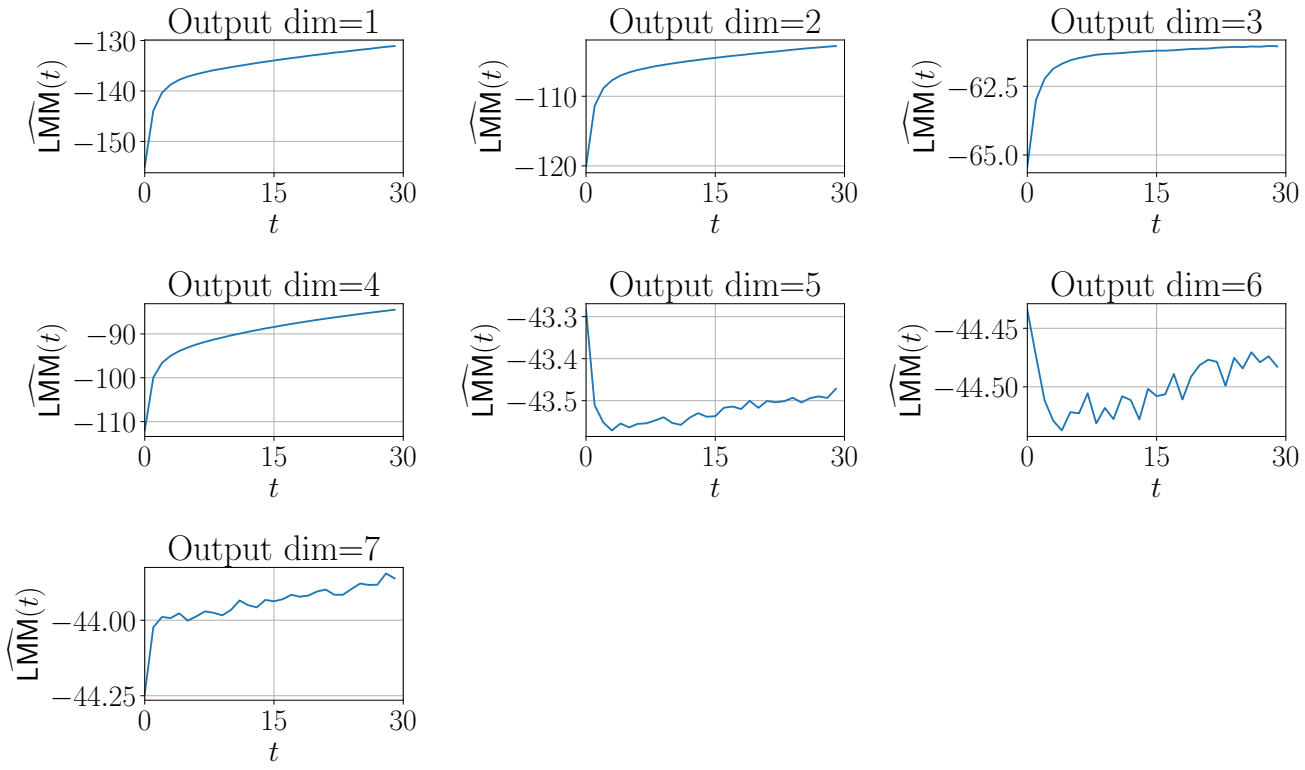


Figure 1: Illustration of $\widehat{\mathsf{LMM}}(t)$ for each output dimension for the SARCOS dataset, during the hyperparameter optimization procedure performed in Section 5.1.

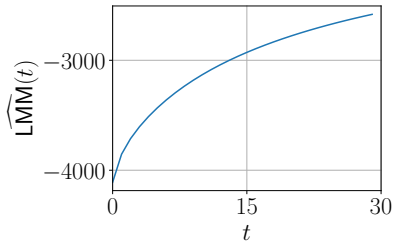


Figure 2: Illustration of $\widehat{\mathsf{LMM}}(t)$ for the Diabetes dataset, during the hyperparameter optimization procedure performed in Section 5.1.